1.0 FLASH INTERFACE (FLI)

1.1 Features

1. Flash read/write with ECC generation
2. Instruction prefetch to improve code execution time
3. Flash operations: read, page erase and program (burn)
4. Interrupt generation for flash operation and ECC errors

1.2 Introduction

In this document the word "burn" is used to mean a "program" operation so as not to be confused with software. A simplified block diagram of FLI is shown in Figure 1. The burn, erase and read operation are managed through the flash control. The burn and erase operation are managed through APB bus and instruction reads are done through AHB bus. It is important to note, it is not possible to execute from flash and do the burning at the same time. The ECC blocks generate and decodes ECC bits if EN_ECC bit is set in flash control register. The flash pre-fetch buffer if enabled, buffer 4 words if the execution of instruction is sequential and can help minimize CPU burden and improve execution speed.

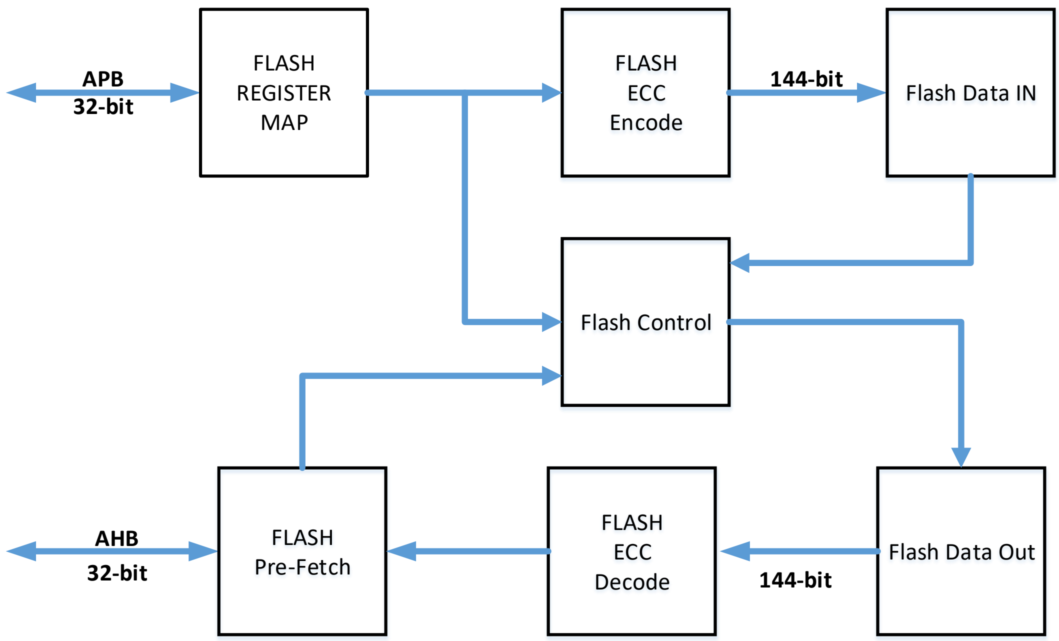

Figure 1: Flash Control Block Diagram

1.3 Flash Memory Map:

The flash memory map is shown in Figure 2. In total there are 64 pages (depending on product selection). Pages 54 and 63 are reserved and protected and cannot be erased or programmed. Flash memory AHB bus access starts at address 0x08000000, ends at address 0x0803FFFF and each flash page is 4kB. Therefore, the flash memory address range for page 0 is from 0x08000000 to 0x08000FFF, page 1 is from 0x08001000 to 0x08001FFF,etc. Flash pages programming is done through programming flash cells. Each flash page contains 256 flash cells and each flash cell contains 128bits of user data (144 bits including ECC bits). Therefore, with each programming operation it is possible to program 128bit of user data. For example, to program address range 0x08000000 to 0x0800000F page 0 (pageaddr = 0), Row (rowaddr = 0) 0 and Column 0 (coladdr = 0) must be selected in the register setting. Detailed instruction for programming the flash is provided in the following section.

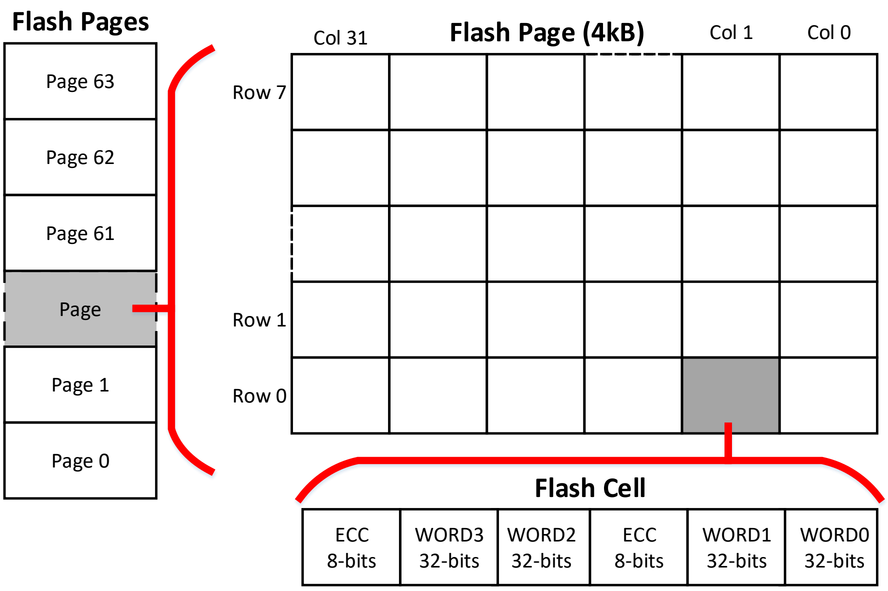

Figure 2: Flash Memory Map

1.4 Flash Memory Erase Operation:

It is possible to erase one flash page at a time. To erase a page, the following sequence must be followed:

1. Configure the flash time, FLTIME, register parameters with correct parameters. 20 µs is required for burn operation and 130 ms is required for erase operation. Assuming the default clock speed of 40MHz is used:
   - FLMS (Flash MicroSecond) = 40 (corresponds to 1µs)
   - FLPRT(Flash Program Time) = 20 (corresponds to 20µs)
   - FLERT(Flash Erase Time) = 130 (corresponds to 130ms)

   If MCU clock is changed to a value other than default parameter, the parameter FLMS must be adjusted accordingly. For example, if MCU clock of 20MHz is used, then FLMS must be set to 80.
2. configure FLMode in FLCTRL register for erase operation
3. Choose desired page to be erased by configuring pageaddr in FLADDR register.
4. Set exe bit in FLEXE register to 1. (note FLEXE is reverted back to 0 automatically when operation is complete.)
5. Wait at least 130ms or until fexecomplete is set in Flashintstat register.

1.5 Flash Memory Program Operation:

It is possible to program one flash page cell at a time which is done in groups of 36 bits (32-bit of user data and 4 bits of ECC if enabled). To program a page cell, the following sequence must be followed:

1. Configure the flash time, FLTIME, register parameters with correct parameters. 20 µs is required for burn operation and 130ms is needed for erase operation. Assuming the default clock speed of 40MHz is used:
   - FLMS (Flash MicroSecond) = 40 (corresponds to 1µs)
   - FLPRT(Flash Program Time) = 20 (corresponds to 20µs)
   - FLERT(Flash Erase Time) = 130 (corresponds to 130ms)

   If MCU clock is changed to a value other than default parameter, the parameter FLMS must be adjusted accordingly. For example, if MCU clock of 20MHz is used, then FLMS must be set to 80.
2. Configure FLMode in FLCTRL register for burn operation. Also set EN_ECC bit to 1, if ECC generation and error detection during flash read operation is required.
3. Configure the data to be written to flash in flashdata (FLDATA) registers 0, 1,2 and 3.
4. Choose desired cell to be programmed by configuring pageaddr, rowaddr,coladdr and group in FLADDR register. (start with group 0 which programs the data in FLDATA register 0 into the first 32 bits of the desired address).
5. Set exe bit in FLEXE register to 1. (note FLEXE is reverted back to 0 automatically when operation is complete.)
6. Wait at least 20µs or until execomplete is set in Flashintstat register.
7. Repeat steps 4-6 for the other groups 1,2 and 3 until the whole flash cell burn is complete (144-bits including ECC bits)
8. Repeat steps 3-6 for the following flash cells.

   Note: Previously burnt bits in a flash cell must be erased first before the next burn operation. If the same bits are burnt twice, unexpected behavior may occur.

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1.6 Flash Memory Read Operation:

Flash read operations require setting the FLMode in the FLCTRL register to 'read'. The pre-fetch buffer impacts read performance. Without it, each read takes two clock cycles. When enabled, the pre-fetch buffer adds a one-cycle wait state for the initial instruction fetch, but subsequent sequential instructions (up to 128 bits) are buffered for zero wait state access. This optimization reduces the time required for four consecutive reads from eight clock cycles down to five, potentially boosting CPU performance.

1.7 Flash Memory Status Information:

Flash status information is available in the Flashintstat register. The execomplete flag within this register signals the completion of erase or burn operations. With ECC enabled, the system reads 144 bits from the flash. The ECC decoder corrects any single-bit errors, indicated by the leccs (for least significant bytes) or meccs (for most significants bytes) flag, and provides the corrected 128-bit data to the data path. Multiple-bit errors are signaled by the leccd or meccd flag. To ensure data integrity, ECC must be enabled during both burn operations (for ECC generation) and read operations (for ECC correction).

1.8 Interrupt Generation:

It is possible to generate interrupt for flash status information as shown in Figure 3. To enable interrupt generation, set the corresponding bit in Flashintenstat register. The flash control interrupt output is connected to system control unit NVIC vector. The flash controller peripheral by itself is not connected to NVIC vector. For more information about generating interrupt in system control unit refer to SCU documentation.

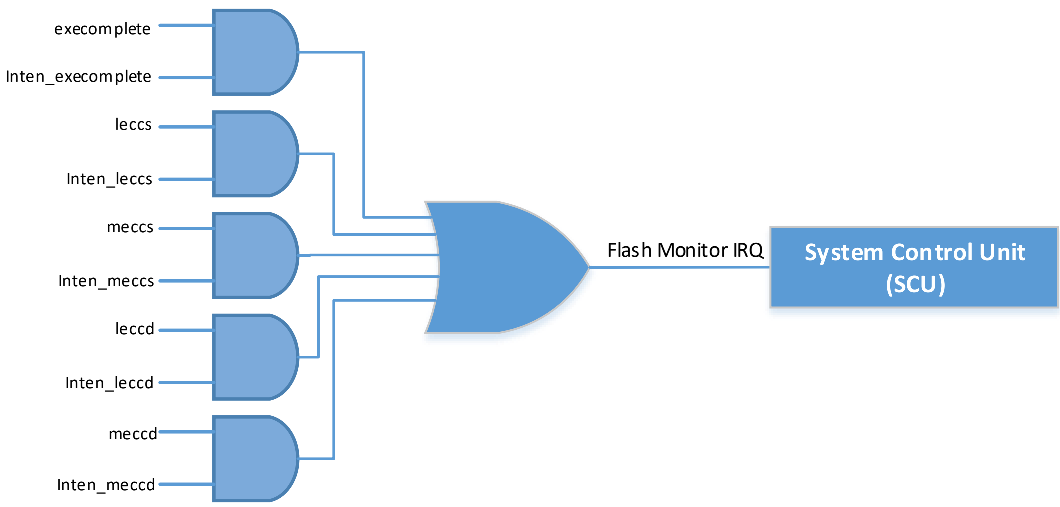

Figure 3: Flash Status Interrupt Signal

1.9 Register sets

The flash controller contains 10 registers in total. The register names, address offsets and access type and reset values are shown in Table 1.

Flash Control Base Address: 0x4000F000

Table 1: Flash Control Registers

Register Name	Address Offset	Access Type	Reset Value
Flash Control (FLCTRL)	0x00	R/W	0x00000000
Flash Time (FLTIME)	0x04	R/W	0x00000000
Flash Address (FLADDR)	0x08	R/W	0x00000000
Flash Execution (FLEXE)	0x0C	R/W	0x00000000
Flash Data 0 (FLDATA0)	0x10	R/W	0x00000000
Flash Data 1 (FLDATA1)	0x14	R/W	0x00000000
Flash Data 2 (FLDATA2)	0x18	R/W	0x00000000
Flash Data 3 (FLDATA3)	0x1C	R/W	0x00000000
Flash Status (Flashintstat)	0x20	R/W	0x00000000
Flash Status Interrupt Enable (Flashintenstat)	0x24	R/W	0x00000000

Flash Control (FLCTRL) Register Address: 0x4000F000

Reset Value: 0x00000000

This register defines flash control, ECC and prefetch mode.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
-							
U	U	U	U	U	U	U	U

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
-							
U	U	U	U	U	U	U	U

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-							
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	EN_PreFetch	EN_ECC	Res	FLMode			
R/W	R/W	R/W	Res	R/W	R/W	R/W	R/W

Legend: R: Read W: Write 1C: Write 1 to clear	U: Unimplemented bit, read as 0 Res: Reserved bits can be R/W but should be kept as default register value. Changing value can result in unexpected behavior.

Field	Bits	Description
FLMode	3:0	Flash operation mode 0x0: Read 0x3: Page Erase 0x4: Program (Burn) All other combinations are reserved and should not be used.
EN_ECC	5	ECC Enable 0x0: Disabled 0x1: Enabled Note1: When enabled, during burn operation ECC bits are generated and programmed to flash. During read operation, ECC bits are decoded and in case of mismatch Flashintstat register is updated with the corresponding error. Note2: During programming the EN_ECC bit must be set high if it is desired to read flash with EN_ECC bit set high. Otherwise code execution during read operation may not work as expected.
EN_PreFetch	6	Flash Instruction Prefetch 0x0: Disabled 0x1:Enabled Note: When this option is enabled, instruction fetch incurs a one cycle-wait state, however, subsequent instructions, if sequential (up to 128-bit of user

data) are buffered and can be read with zero wait state. This option can improve CPU execution speed.

Flash Time (FLTIME) Register Address: 0x4000F004

Reset Value: 0x00000000

This register defines the timing parameters needed for the flash page erase and program operations.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
-							
U	U	U	U	U	U	U	U

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
FLERT							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-			FLPRT				
U	U	U	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
FLMS							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bits can be R/W but should be kept as default
1C: Write 1 to clear	register value. Changing value can result in unexpected behavior.

Field	Bits	Description
FLMS	7:0	Flash micro seconds $micro\ second\ Time = FLMS \times CPU\ Clock\ Speed$ The FLMS parameter needs to be set such that the time is equal to 1 µs. With a default CPU clock speed of 40 MHz, set FLMS to 40.
FLPRT	12:8	Flash program time $Flash\ Program\ Time = FLPRT \times 1\ \mu s$ Note: A flash program time of 20 us is required. Set FLPRT to 20.
FLERT	23:16	Flash Erase Time $Flash\ Page\ Erase\ Time = FLERT \times 1\ ms$ Note: Flash page erase time of 130 ms is required. Set FLERT to 130.

Flash Address (FLADDR) Register Address: 0x4000F008

Reset Value: 0x00000000

Defines the flash cell that needs to be programmed.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
-							
U	U	U	U	U	U	U	U

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
-						group	
U	U	U	U	U	U	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	pageaddr					
RES	RES	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
rowaddr			coladdr				
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend: R: Read W: Write 1C: Write 1 to clear	U: Unimplemented bit, read as 0 Res: Reserved bits can be R/W but should be kept as default register value. Changing value can result in unexpected behavior.

Field	Bits	Description
coladdr	4:0	Flash page column address Check flash memory map for more information.
rowaddr	7:5	Flash page row address Check flash memory map for more information.
pageaddr	13:8	Flash Page Address Check flash memory map for more information.
group	17:16	Data group This parameter decides the data group to be burnt into flash in the next program operation. 0x0: data in flash data 0 (FLDATA0) register is burnt into flash in the next execution cycle 0x1: data in flash data 1 (FLDATA1) register is burnt into flash in the next execution cycle 0x2: data in flash data 2 (FLDATA2) register is burnt into flash in the next execution cycle 0x3: data in flash data 3 (FLDATA3) register is burnt into flash in the next execution cycle The program execution needs to start from group 0 and end with group 3. Check Flash Memory Program Operation section for more information.

Flash Execution (FLEXE) Register Address: 0x4000F00C

Reset Value: 0x00000000

This register executes the flash burn and erase operation

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
-							
U	U	U	U	U	U	U	U

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
-							
U	U	U	U	U	U	U	U

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-							
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-							exe
U	U	U	U	U	U	U	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bits can be R/W but should be kept as default
1C: Write 1 to clear	register value. Changing value can result in unexpected behavior.

Field	Bits	Description
exe	0	Flash execute operation When set to 1, flash burn or program operation which is configured by FLMode parameter starts. Note 1: execomplete bit in register Flashintstat will be set when the execution is complete. Note 2: The exe bit automatically clears to 0 upon operation completion.

Flash Data (FLDATA)

FLDATA0 Register Address: 0x4000F010
FLDATA1 Register Address: 0x4000F014
FLDATA2 Register Address: 0x4000F018
FLDATA3 Register Address: 0x4000F01C

Reset Value: 0x00000000

This register contains the data that is used to program the flash.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
DATA[31:24]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
DATA[23:16]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
DATA[15:8]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
DATA[7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bits can be R/W but should be kept as default
1C: Write 1 to clear	register value. Changing value can result in unexpected behavior.

Field	Bits	Description
DATA	31:0	Flash data This register defines the 32-bit user data that needs to be programmed into the flash.

Flash Status (Flashintstat) Register Address: 0x4000F020

Reset Value: 0x00000000

This register provides status information about flash operation status and ECC.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
-							
U	U	U	U	U	U	U	U

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
-							
U	U	U	U	U	U	U	U

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-							
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	Reserved		meccd	meccs	leccd	leccs	execomplete
U	Res	Res	R/1C	R/1C	R/1C	R/1C	R/1C

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bits can be R/W but should be kept as default
1C: Write 1 to clear	register value. Changing value can result in unexpected behavior.

Field	Bits	Description
execomplete	0	Flash execution complete bit 0: execution not started or complete 1: execution is complete
leccs	1	Least significant byte ECC single error 0: no error 1: ECC single error
leccd	2	Least significant byte ECC double error 0: no error 1: ECC double error
meccs	3	Most significant byte ECC single error 0: no error 1: ECC single error
meccd	4	Most significant byte ECC double error 0: no error 1: ECC double error

Flash Status (Flashintenstat) Register Address: 0x4000F024

Reset Value: 0x00000000

This register enables interrupt generation for flash status register.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
-							
U	U	U	U	U	U	U	U

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
-							
U	U	U	U	U	U	U	U

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-							
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	Reserved		Inten_meccd	Inten_meccs	Inten_leccd	Inten_leccs	Inten_execomplete
U	Res	Res	R/W	R/W	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bits can be R/W but should be kept as default
1C: Write 1 to clear	register value. Changing value can result in unexpected behavior.

Field	Bits	Description
Inten_execomplete	0	Enables interrupt generation for execution complete bit 0: disabled 1: enabled
Inten_leccs	1	Enables interrupt generation for least significant byte ECC single error 0: disabled 1: enabled
Inten_leccd	2	Enables interrupt generation for least significant byte ECC double error 0: disabled 1: enabled
Inten_meccs	3	Enables interrupt generation for most significant byte ECC single error 0: disabled 1: enabled
Inten_meccd	4	Enables interrupt generation for most significant byte ECC double error 0: disabled 1: enabled

Note: The NVIC vector for interrupt generation for this peripheral is connected to system control unit and there is no standalone vector for this peripheral. Check system control unit (SCU) document for more information about interrupt generation for flash status information.

Revision Control:

20/03/2025	Initial Release
08/07/2025	Added additional note for EN_ECC bit in the register map table.

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